

FEATURES

Nutrition Assistant System

The **Nutrition Assistant** is a full-stack MERN web application that provides personalized nutrition guidance based on the user's health information. It helps users calculate BMI, generate diet plans, monitor nutrition history, and allows administrators to manage users and nutrition suggestions efficiently.

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Nutrition Assistant System

The Nutrition Assistant is a full-stack MERN web application that provides personalized nutrition guidance based on the user's health information. It helps users calculate BMI, generate diet plans, monitor nutrition history, and allows administrators to manage users and nutrition suggestions efficiently.

KEY FEATURES

1



User Registration and Authentication
Users can create an account securely using their personal details. Passwords are encrypted using bcrypt and authentication is handled through JWT to ensure secure access.

2



Personalized Nutrition Plan Creation
Users enter age, height, weight, gender, and activity level. Based on this data, the system generates a personalized nutrition plan with calorie intake, protein requirements, carbohydrate needs, meal timing, and walking recommendations.

3



BMI Calculation
The application automatically calculates the user's Body Mass Index (BMI). Users are categorized as Underweight, Normal Weight, Overweight, or Obese based on their BMI value.

4



Nutrition Recommendation Generation
The backend generates personalized nutrition suggestions including daily calorie requirement, protein intake, carbohydrate requirement, healthy foods, meal schedule, walking recommendation, and weight management guidance.

5



Nutrition History
All generated nutrition plans are stored securely in MongoDB. Users can access their previous nutrition plans and monitor their health progress over time.

6



User Dashboard
After logging in, users are redirected to their personalized dashboard where they can view profile details, create new diet plans, view previous nutrition suggestions, track their nutrition history, and manage their account information.

7



Responsive User Interface
The frontend is developed using React.js and Bootstrap 5, providing a modern, responsive, and user-friendly interface that works seamlessly across desktops, laptops, tablets, and mobile devices.

8



Admin Dashboard
The Admin Dashboard provides centralized management features, allowing admins to view all registered users, monitor nutrition suggestions, delete inappropriate or outdated suggestions, manage user activities, and maintain system records.

9



Secure REST API Integration
The application follows RESTful architecture for communication between frontend and backend. All API requests are validated and protected using authentication middleware, ensuring secure and reliable data exchange.

10



Database Management
MongoDB is used to store user information and nutrition suggestions. Mongoose schemas ensure proper data validation and maintain relationships between Users and Suggestions, providing efficient storage and retrieval of application data.

1. User Registration and Authentication

The system allows new users to create an account securely using their personal details. Passwords are encrypted using bcrypt, and authentication is handled through JWT (JSON Web Token) to ensure secure access to protected resources.

2. Personalized Nutrition Plan Creation

Users can enter health information such as age, height, weight, gender, and activity level. Based on this data, the application generates a personalized nutrition plan with calorie intake, protein requirements, carbohydrate needs, meal timing, and walking recommendations.

3. BMI Calculation

The application automatically calculates the user's Body Mass Index (BMI) using height and weight. Based on the BMI value, the user is categorized as:

- Underweight
- Normal Weight
- Overweight
- Obese

This classification is used to generate suitable nutrition recommendations.

4. Nutrition Recommendation Generation

The backend processes the user's health details and generates personalized nutrition suggestions, including:

- Daily calorie requirement
- Protein intake
- Carbohydrate requirement
- Recommended healthy foods
- Meal schedule
- Walking recommendation
- Weight management guidance

5. Nutrition History

Every generated nutrition plan is stored securely in MongoDB. Users can access their previous nutrition plans and monitor their health progress over time.

6. User Dashboard

After logging in, users are redirected to a personalized dashboard where they can:

- View profile details
- Create new diet plans
- View previous nutrition suggestions
- Track their nutrition history
- Manage their account information

7. Responsive User Interface

The frontend is developed using React.js and Bootstrap 5, providing a modern, responsive, and user-friendly interface that works seamlessly across desktops, laptops, tablets, and mobile devices.

8. Admin Dashboard

The Admin Dashboard provides centralized management features, allowing administrators to:

- View all registered users
- Monitor nutrition suggestions
- Delete inappropriate or outdated suggestions
- Manage user activities
- Maintain system records

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The application follows RESTful architecture for communication between the frontend and backend. All API requests are validated and protected using authentication middleware, ensuring secure and reliable data exchange.

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